Teacher guide: Lesson 2, Make It Yours: Light Patterns

For: learners aged 10–15 (or adult beginners) who have built the Lesson 1 traffic light. **New idea:** give things names: variables (`RED_S = 5`), one function (`set_lights`) and `print()` to see what the program is doing. **New part:** none. The circuit is exactly the Lesson 1 circuit. **Time:** about 40 minutes. **Needs:** the kit, one shared laptop with Thonny (works offline once installed) and the printed worksheet. No phones.

> This is a low-voltage circuit (3.3 V from the ESP32, powered by USB). Keep the USB cable **unplugged while checking or changing wires**, and have an adult check every circuit before it is plugged in. The lesson's 3D animation is a learning picture, not a safety test.

1. Before class (10 minutes, once per board)

1. The board needs MicroPython already on it (as for Lesson 1).
2. Copy `lessons/02-patterns/main.py` onto each board as **`main.py`**, replacing the Lesson 1 program (in Thonny: open the file, *File → Save as... → MicroPython device*, name it `main.py`). Keep a copy on the laptop too.
3. Check the board: unplug, plug back in. Red should light within a second or two, then green, then yellow.
4. Use a **data** USB cable. Some cables only charge and cannot load programs or show messages.

The lights do **exactly the same as in Lesson 1** (red 5 s → green 5 s → yellow 2 s, a 12 s cycle). That is the point: the program is easier to read and change, but it does the same job.

2. Timing (40 minutes)

Minutes	What happens
0–5	Recap: "Last time you built a traffic light." Hand out the worksheet. Show the program in Thonny on the laptop (or read section 8 aloud). Learners do **1. Predict**.
5–10	**2. Check the circuit** with USB unplugged (section 4), then plug in.
10–18	Variables: `RED_S = 5` stores 5 in a box labelled `RED_S`. `=` means "store", not "equals". Learners count the seconds (**3. Observe**, first line).
18–25	The function: `def` teaches the ESP32 a new word; `set_lights(1, 0, 0)` fills in r, y, g. Learners answer the `set_lights` rows in **1. Predict**.
25–30	`print()`: connect the laptop, press Run in Thonny (section 5). Learners copy three Shell messages (**3. Observe**).
30–38	**4. Make it yours**: each group designs a pattern on paper; you or a learner types it in and runs it (section 6).
38–40	**5. Explain** and discussion questions (section 7). Put the class `main.py` back if anyone saved over it.

3. Parts list (per board or pair of learners)

The Lesson 1 traffic light, still built: ESP32 DevKit V1 with MicroPython, breadboard, red, yellow and green LEDs, three 330 Ω resistors, three jumper wires to D25, D26 and D27, three short black wires and one long black GND wire. Plus: one USB **data** cable and one shared laptop with Thonny for the class (or one per group, if you have them).

4. Circuit check (USB unplugged; same as Lesson 1)

Light	ESP32 pin → hole	330 Ω resistor	Long leg (+)	Short leg (−)	Black wire
Red	D25 → b7	d7 → g7 (across the gap)	i7	i8	j8 → − rail
Yellow	D26 → b15	d15 → g15	i15	i16	j16 → − rail
Green	D27 → b23	d23 → g23	i23	i24	j24 → − rail
Ground	GND pin next to D13 → blue − rail				

Read the pin labels; don't count pins. If a circuit was taken apart since Lesson 1, use the full 8-step check in the Lesson 1 teacher guide (section 5). Nothing new is wired today, and no other ESP32 pin is used.

5. Seeing `print()` messages on the shared laptop

1. Plug the board into the laptop with the data cable. Open Thonny. Bottom right, choose *MicroPython (ESP32)* and the board's port.
2. Thonny stops the running program when it connects, so the lights may freeze. That is normal.
3. Open `main.py` (from the laptop copy or from the device) and press the green **Run** button (or F5).
4. The lights start again and the **Shell** under the code shows `Red: stop`, `Green: go`, `Yellow: get ready`, one message each time the light changes. Press the red **Stop** button to stop.
5. Unplug and plug back in: the board runs the `main.py` saved on it again, with no laptop needed.

`print()` never changes the lights. It only sends text down the USB cable. Without the laptop the messages go nowhere.

6. Make it yours: the three patterns from the lesson page

To try a pattern, change only the lines shown below in the laptop copy, press **Run**, and watch. Keep the top of the program (`from machine…`, the three `Pin` lines and `def set_lights`) as it is. **Run** does not save anything on the board; to keep a pattern after unplugging, save it on the device as `main.py`.

Pattern	Variable lines	Lines inside `while True:` (each indented 4 spaces)
Classic (the class program)	`RED_S = 5`, `GREEN_S = 5`, `YELLOW_S = 2`	see section 8
Warning flasher	`FLASH_S = 0.5`	`print("Flash on")`, `set_lights(0, 1, 0)`, `time.sleep(FLASH_S)`, then `print("Flash off")`, `set_lights(0, 0, 0)`, `time.sleep(FLASH_S)`
Light chase	`STEP_S = 1`	`print("Red")`, `set_lights(1, 0, 0)`, `time.sleep(STEP_S)`, then the same three lines for `Yellow` `(0, 1, 0)` and `Green` `(0, 0, 1)`

When you replace the variable lines, also replace the loop lines, or Python stops with a NameError (for example `RED_S` is used but no longer set). Challenges from the page: red for 8 s (change only `RED_S = 5` to `RED_S = 8`); the flasher twice as fast (`FLASH_S = 0.25`); the chase in the order green, yellow, red (swap the numbers in the first and last `set_lights`); then the learners' own pattern from worksheet section 4, each phase three lines: `print`, `set_lights`, `time.sleep`.

7. Likely problems (and fixes)

What you see	Most likely cause	Fix
`NameError: name 'red_s' isn't defined`	Spelling or capitals differ: `red_s` is not `RED_S`	Copy the name exactly as it is written at the top.
`NameError` for `set_lights`	The loop was typed above `def set_lights`, or `def` was deleted	Python reads top to bottom: `def` must come before `while True:`.
`IndentationError`	Lines inside `def` or `while True:` are not lined up	Use exactly 4 spaces before each line inside them.
`SyntaxError`	A missing comma or bracket, or a message without both quote marks	Compare with section 8: `set_lights(1, 0, 0)` and `print("Red: stop")`.
The wrong colour lights up	The three numbers are in the wrong order	The order is always red, yellow, green, as in `def set_lights(r, y, g)`.
Nothing appears in the Shell	Thonny is not connected, the wrong port is chosen, or a charge-only cable	Check the port (bottom right), press Stop then Run, try another cable.
After unplugging, the board does the old pattern	The new pattern was only Run, not saved	Save on the device as `main.py`, or leave the class program there on purpose.
No light at all, even with the class program	The circuit has changed since Lesson 1	Do the check in section 4.

8. The program (for reference; the same as the lesson page with its default times, plus one comment)

```
# Light Patterns (ESP32)
from machine import Pin
import time

red = Pin(25, Pin.OUT)
yellow = Pin(26, Pin.OUT)
green = Pin(27, Pin.OUT)

# How long each light stays on (seconds)
RED_S = 5
GREEN_S = 5
YELLOW_S = 2

def set_lights(r, y, g):
    red.value(r)
    yellow.value(y)
    green.value(g)

# print() messages appear in the Shell in Thonny when the board is connected to the laptop.
while True:
    print("Red: stop")
    set_lights(1, 0, 0)
    time.sleep(RED_S)

    print("Green: go")
    set_lights(0, 0, 1)
    time.sleep(GREEN_S)

    print("Yellow: get ready")
    set_lights(0, 1, 0)
    time.sleep(YELLOW_S)
```

9. Discussion questions and worksheet answers

1. Why `time.sleep(RED_S)` and not `time.sleep(5)`? (The name says what the number means, and you change it in one place.)
2. Why does `set_lights` switch all three lights every time? (Nothing from before stays on; red and green are never on together.)
3. Why must `def` come before the loop? (Python reads top to bottom; otherwise it does not know the word yet.)
4. Where did the `print()` messages go? (Down the USB cable to the Shell in Thonny on the laptop.)
5. Lesson 1 needed three `value()` lines per phase. How many lines does a phase need now? (Lesson 1 needed four, three of them to switch lights. Now three, and only `set_lights` switches lights.)

Worksheet: 1. No, same lights and times; yellow; none, all off; 8 + 5 + 2 = 15 s. 3. 5, 5, 2 s, yes; `Red: stop`, `Green: go`, `Yellow: get ready`.